

25CSCI20H

Advanced Human-Computer Interaction Python Basics

From installation → first programs → importing libraries

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ANACONDA.

What we'll cover

- What is Anaconda (Distribution + conda)?
- What is Anaconda Navigator (GUI) and why it's useful
- Installation overview
- What is Jupyter Notebook (and JupyterLab)?
- Python basics: variables, I/O, comments, if, loops, functions
- Libraries: importing, installing, and environment hygiene

Goal by the end

- You can install Anaconda and open Jupyter
- You can create a project environment
- You can write small Python programs confidently

What is Anaconda?



Think of Anaconda Distribution as a “starter kit” for Python + data science.

- Includes Python + the conda package/environment manager
- Ships with a large collection of popular packages (NumPy, pandas, etc.)
- Bundled tools include Anaconda Navigator (GUI) and often Jupyter
- Designed to reduce “dependency conflicts” by using isolated environments

Why people use it

- Easy install (especially for beginners)
- Manage packages without pain
- Switch project setups safely

Conda environments (the “secret sauce”)

An environment is an isolated set of packages + (optionally) a specific Python version.

base (default)

- Python + conda
- Common tools
- Keep it clean

project: “ml101”

- python=3.11
- numpy, pandas
- matplotlib

project: “web-api”

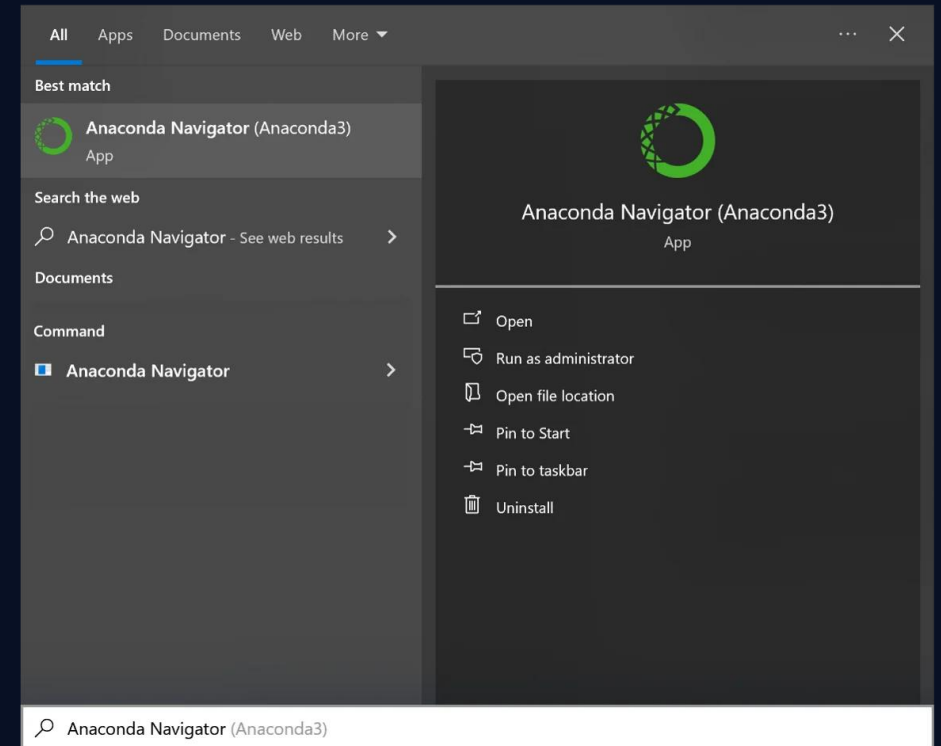
- python=3.12
- fastapi, uvicorn
- requests

Why it matters: two projects can need different versions — environments prevent conflicts.

What is Anaconda Navigator?

A desktop GUI included with Anaconda Distribution for managing environments + launching tools.

- Create/activate environments (no terminal required)
- Install/update packages (and channels)
- Launch Jupyter, Spyder, VS Code, etc. from one place



Find & open Navigator

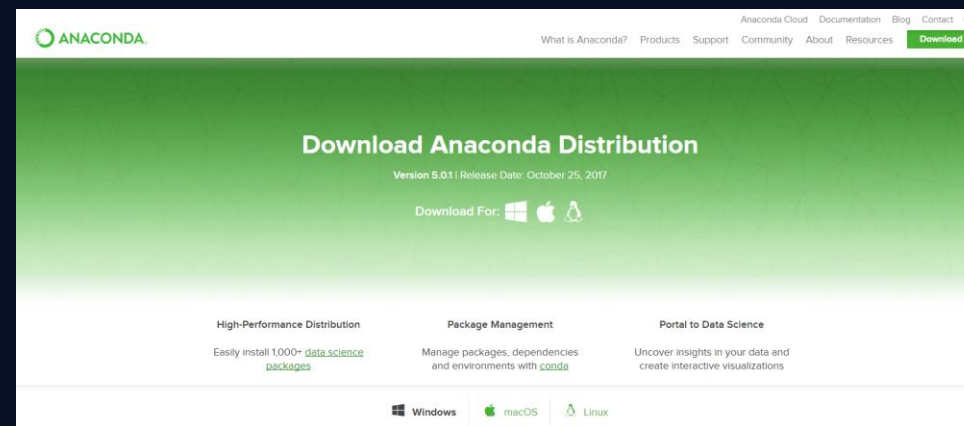
Installing Anaconda Distribution (quick overview)

Install steps

- Download the installer for your OS from anaconda.com
- Run the installer and accept defaults (usually “Just Me”)
- Open Navigator (GUI) or Anaconda Prompt/Terminal (CLI)
- Verify: run Python and/or launch Jupyter

Two beginner-friendly tips

- Prefer a simple install path
- Don’t “dump everything” into base — make environments



Download page (example)

Create an environment + install packages

You can do this in Anaconda Prompt/Terminal (CLI) or via Navigator (GUI).

CLI (recommended to learn)

```
1 # Create an environment (pick a name you like)
2 conda create -n myenv python=3.11
3
4 # Activate it
5 conda activate myenv
6
7 # Install packages
8 conda install numpy pandas matplotlib
9
10 # List what you have
11 conda list
```

Navigator clicks (roughly)

- Environments → Create → choose Python version
- Select env → Install packages
- Home → Launch Jupyter / Spyder

Best practices

- Name envs after projects: “ml101”, “web-api”
- Pin Python version for reproducibility
- Prefer conda packages first; use pip if needed

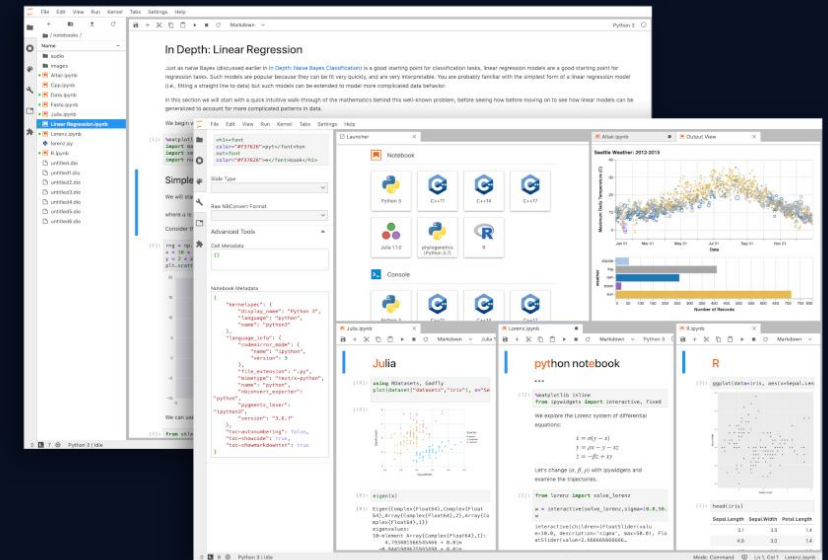
What is Jupyter Notebook (and JupyterLab)?

A notebook lets you mix live code, notes (Markdown), and outputs (tables/plots) in one document.

- Great for learning, experimentation, data analysis, and sharing results
- Run cells step-by-step (instead of running a whole file)
- Notebook files are saved as .ipynb

How you open it

- Navigator → Home → Launch Jupyter
- Or terminal: `jupyter lab / jupyter notebook`



JupyterLab interface (example)

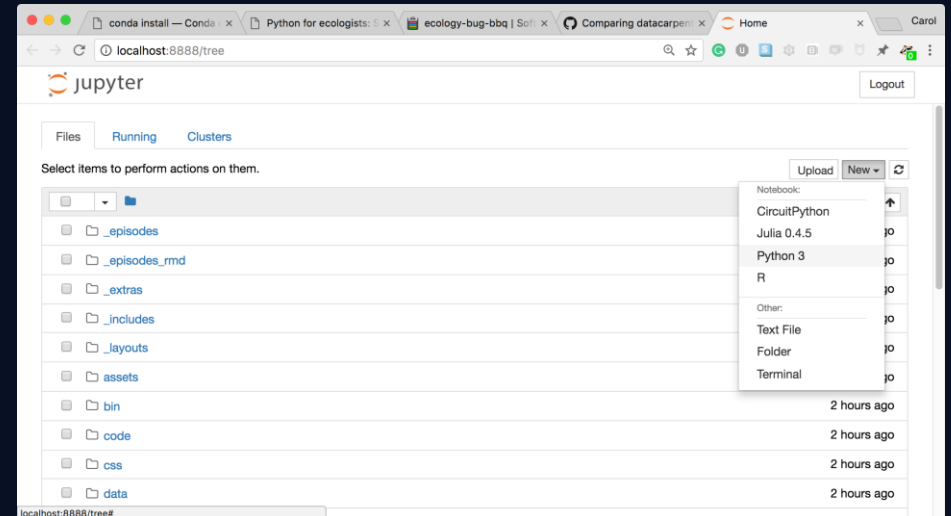
Jupyter basics (the 90-second tour)

Notebook anatomy

- A notebook is made of cells (Code or Markdown)
- A kernel runs your code and returns output
- Outputs can be text, tables, charts, images

Handy shortcuts

- Run cell: Shift + Enter
- Add cell below: B • above: A
- Command palette: Ctrl/Cmd + Shift + P



Notebook dashboard (example)

Python basics: variables & types

Variables are names that point to values. Python figures out the type automatically.

Try this in a notebook cell

```
1 name = "Hassan"
2 age = 24
3 height_m = 1.78
4
5 is_student = True
6
7 print(f"{name} is {age} years old")
8 print(type(height_m), type(is_student))
```

Common types

- int (whole numbers)
- float (decimals)
- str (text)
- bool (True/False)
- list / dict (collections)

Python basics: input/output & comments

I/O essentials

- `print(...)` shows output
- `input(...)` reads text from the user
- Convert types when needed: `int(...)`, `float(...)`

Comments start with #

```
1 # Ask the user for a number and double it
2
3 n = int(input("Enter a number: "))
4 d = n * 2 # multiply by 2
5 print("Double =", d)
```

Mini practice

- Ask for a name, then say hello
- Ask for two numbers, print their sum
- Ask for age, print age next year

Python basics: if-statements & loops

Control flow lets your program make decisions and repeat work.

if / elise f / else + for

```
1 score = 72
2
3 if score >= 70:
4     letter = "A"
5 elif score >= 60:
6     letter = "B"
7 else:
8     letter = "C"
9
10 for i in range(3):
11     print(i, letter)
```

while

```
1 # while loop example
2 count = 3
3 while count > 0:
4     print("count =", count)
5     count -= 1
6 print("Done")
```

Common patterns

- Loop over a list: for item in items
- Count loops: for i in range(n)
- Validate input: while not ok: ...

Python basics: functions & libraries

Functions

- Bundle logic into reusable blocks
- Inputs are parameters; output is returned
- Keeps code clean and testable

Importing standard libraries

```
1 import math
2 from datetime import date
3
4 def area_circle(r):
5     return math.pi * r**2
6
7 today = date.today()
8 print("Today:", today)
9 print("Area:", area_circle(3))
```

Installing 3rd-party libraries

```
1 # In Anaconda Prompt /
  Terminal
2 conda install pandas
3
4 # If a package isn't on conda,
  use pip
5 pip install pandas
```

- Always install into the active environment
- Keep a requirements list (export env if needed)

Next steps (practice path)

3 mini-projects to lock it in

- 1) Number guesser (loops + conditionals)
- 2) Grade calculator (functions + lists)
- 3) CSV mini-analysis (pandas + Jupyter)

Cheat sheet commands

- `conda create -n myenv python=3.11`
- `conda activate myenv`
- `conda install <pkg>`
- `jupyter lab`

Good references

https://youtu.be/Ro_MScTDfU4?si=1y1nUvnkpiX-dM3U